

PRACTICE MIDTERM EXAM

1. The probability that an international flight leaving the United States is delayed in departing (event D) is 0.25. The probability that an international flight leaving the United States is a transpacific flight (event P) is 0.57. The probability that an international flight leaving the U.S. is a transpacific flight and is delayed in departing is 0.12.
 - a. What is the probability that an international flight leaving the United States is delayed in departing given that the flight is a transpacific flight?
 - b. In this problem, are D and P independent?
2. A telephone survey conducted by the Maritz Marketing Research company found that 43% of Americans expect to save more money next year than they saved last year. Forty-five percent of those surveyed plan to reduce debt next year. Of those who expect to save more money next year, 81% plan to reduce debt next year. An American is selected randomly.
 - a. What is the probability that this person expects to save more money next year and plans to reduce debt next year?
 - b. What is the probability that this person expects to save more money next year or plans to reduce debt next year?
 - c. What is the probability that this person neither expects to save more money next year nor plans to reduce debt next year?
 - d. What is the probability that this person expects to save more money next year and does not plan to reduce debt next year?
3. A study on ethics in the workplace by the Ethics Resource Center and Kronos, Inc., revealed that 35% of employees admit to keeping quiet when they see coworker misconduct. Suppose 75% of employees who admit to keeping quiet when they see coworker misconduct call in sick when they are well. In addition, suppose that 40% of the employees who call in sick when they are well admit to keeping quiet when they see coworker misconduct. If an employee is randomly selected, determine the following probabilities:
 - a. The employee calls in sick when well and admits to keeping quiet when seeing coworker misconduct.
 - b. The employee admits to keeping quiet when seeing coworker misconduct or calls in sick when well.
 - c. Given that the employee calls in sick when well, he or she does not keep quiet when seeing coworker misconduct.
 - d. The employee neither keeps quiet when seeing coworker misconduct nor calls in sick when well.
 - e. The employee admits to keeping quiet when seeing coworker misconduct and does not call in sick when well.
4. A random variable X has the following probability mass function:

X	0	1	2	3	4
$P(X = x)$	0.41	0.37	0.16	0.05	0.01

- Find the probability distribution function $F(x)$.
- Find $E(X)$ and $\text{var}(X)$.

5. Let X be a continuous random variable with probability density function given by

$$f(x) = \begin{cases} c(x^3 + 2x + 1), & 0 \leq x \leq 4 \\ 0, & \text{otherwise} \end{cases}$$

- Find c .
- Find $P(1 < X)$.
- Find $E(2X + 1)$

6. The joint probability distribution of a pair of random variables is given by the following table:

$Y \backslash X$	1	2	3
1	1/6	1/6	1/6
2	1/6	1/12	1/12
3	1/12	1/12	0

- Find the marginal probability mass functions of X and Y .
- Find the conditional distribution of X given $Y = 1$.
- Find $P(X + Y < 4)$.
- Are X and Y independent?

7. The random variables X and Y have a joint pdf:

$$f(x, y) = \begin{cases} c(3x - y), & 1 \leq x \leq 2; 1 \leq y \leq 3 \\ 0, & \text{otherwise} \end{cases}$$

- Find c
- Find the marginal density functions of the random variables X .
- Determine the conditional probability $P(Y > 2 | X = 1.5)$
- Find $P(X + Y < 3)$.

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